

R for bioinformatics, data visualisation, part 2

HUST Bioinformatics course series

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Section 1

section 1: TOC

前情提要

iterations 与 并行计算

- for loop
- apply functions
- dplyr 的本质是 遍历
- map functions in purrr package
- 遍历 与 并行计算

相关包

- purrr
- parallel
- foreach
- iterators

前情提要, cont.

ggplot2 basics

- 图层 (`geom_XXX`)
- `scale` (`scale_XXX`)
- 坐标系统
- faceting (`facet_XXX`)

本次提要

part 2: ggplot2 进阶

- ① 如何在一张图中画多个 panel?
- ② 如何写公式
- ③ ggplot2 的核心是先计算, 再做图
- ④ themes and legends
- ⑤ stacked bars 及其它

Section 2

ggplot2 进阶 1：如何在一张图中画多个 panel?

key requirements for multi-panel plots

- order / position
- labeling
- layout

combine multiple plots

Useful packages:

- gridExtra
- cowplot
- grid
- lattice

install or load packages

```
if (!require("gridExtra")){  
  install.packages("gridExtra");  
}  
  
if (!require("cowplot")){  
  install.packages("cowplot");  
}  
  
library( cowplot );  
library( gridExtra );
```

arranging multiple graphs using cowplot

cowplot:

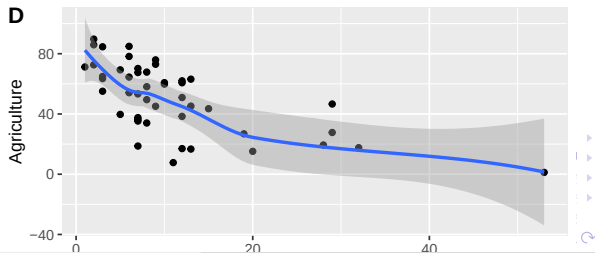
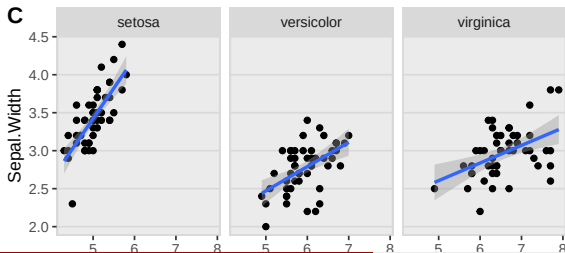
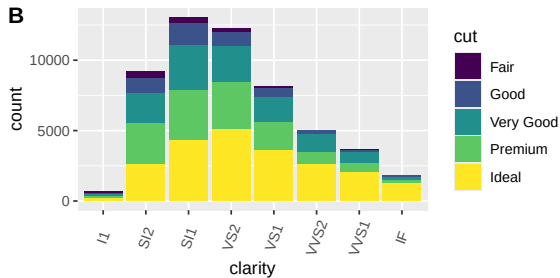
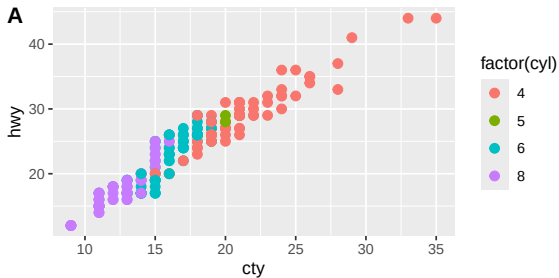
- 1 `plot_grid`
- 2 `ggdraw + draw_plot`

prepare four panels

```
library(tidyverse);
plot.hwy <- ggplot(mpg, aes(x = cty, y = hwy, colour = factor(cyl)))+
  geom_point(size=2.5)
# Bar plot
plot.dose <- ggplot(diamonds, aes(clarity, fill = cut)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle=70, vjust=0.5))
plot.iris <- ggplot(iris, aes(Sepal.Length, Sepal.Width)) +
  geom_point() + facet_grid(. ~ Species) + stat_smooth(method = "lm") +
  background_grid(major = 'y', minor = "none") + # add thin horizontal lines
  panel_border();
plot.swiss <- swiss %>% ggplot(aes(Education, Agriculture)) +
  geom_point() + geom_smooth();
```

Combine four plots using plot_grid

```
cowplot::plot_grid(plot.hwy, plot.dose, plot.iris, plot.swiss,
  labels=c("A", "B", "C", "D"), ncol = 2, nrow = 2)
```

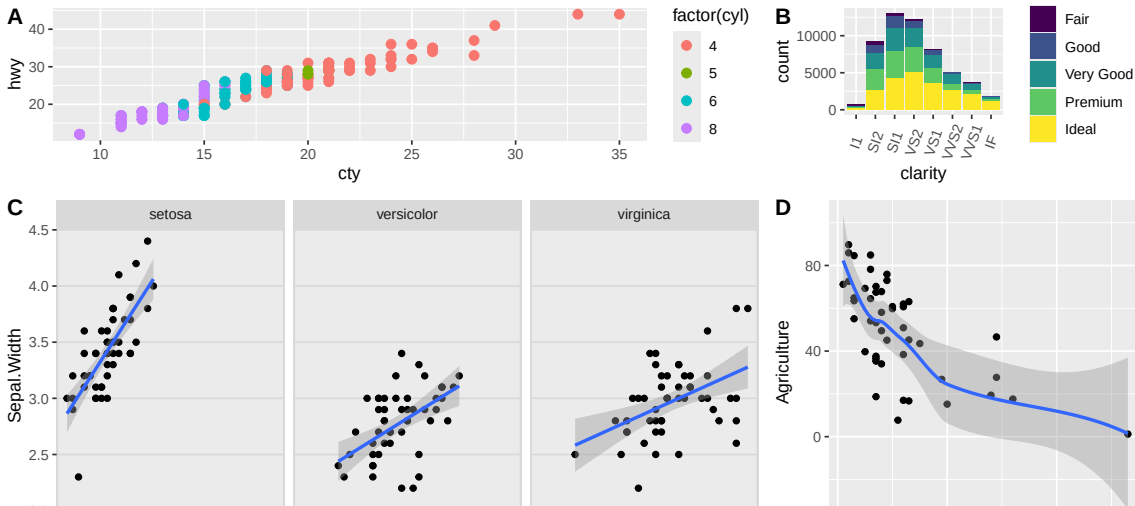


plot_grid parameters

```
plot_grid(
  ...,
  plotlist = NULL,
  align = c("none", "h", "v", "hv"),
  axis = c("none", "l", "r", "t", "b", "lr", "tb", "tblr"),
  nrow = NULL,
  ncol = NULL,
  rel_widths = 1, ## 相对宽度
  rel_heights = 1, ## 相对高度
  labels = NULL,
  label_size = 14,
  label_fontfamily = NULL,
  label_fontface = "bold",
  label_colour = NULL,
  label_x = 0,
  label_y = 1,
  hjust = -0.5,
  vjust = 1.5,
  scale = 1,
  greedy = TRUE,
  byrow = TRUE,
```

调整相对大小

```
cowplot::plot_grid(plot.hwy, plot.dose, plot.iris, plot.swiss,
  labels=c("A", "B", "C", "D"), ncol = 2, nrow = 2,
  rel_widths = c(2,1), rel_heights = c(1, 2));
```



more complex layout using draw_plot

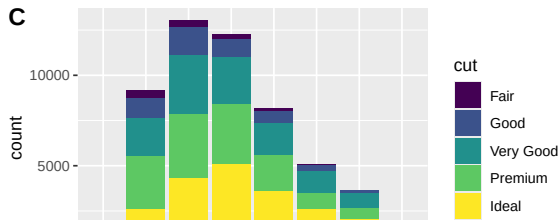
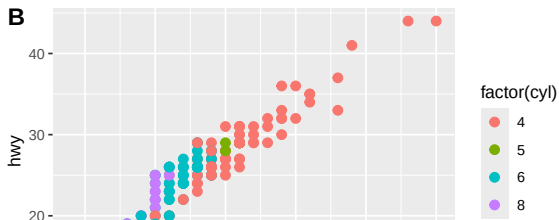
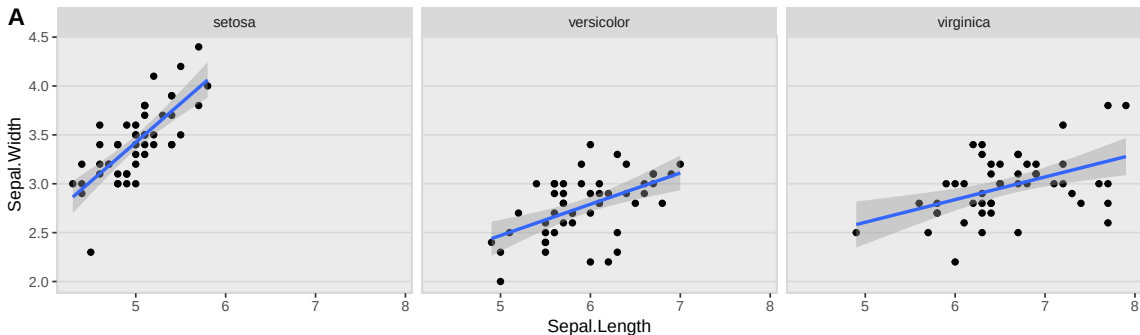
```
plot <-
  ggdraw() +
  draw_plot(plot.iris, x=0, y=.5, width=1, height=.5) +
  draw_plot(plot.hwy, 0, 0, .5, .5) +
  draw_plot(plot.dose, .5, 0, .5, .5) +
  draw_plot_label(c("A", "B", "C"), c(0, 0, 0.5), c(1, 0.5, 0.5), size = 15);
```

`draw_plot(plot, x = 0, y = 0, width = 1, height = 1)` 详解:

- `plot`: the plot to place (ggplot2 or a gtable)
- `x`: The x location of the lower left corner of the plot.
- `y`: The y location of the lower left corner of the plot.
- `width, height`: the width and the height of the plot

draw_plot results

plot



draw_plot_label parameters

Use `draw_plot_label` to add the labels

```
draw_plot_label(c("A", "B", "C"), c(0, 0, 0.5), c(1, 0.5, 0.5), size = 15);
```

```
draw_plot_label(  
  label,  
  x = 0,  
  y = 1,  
  hjust = -0.5,  
  vjust = 1.5,  
  size = 16,  
  fontface = "bold",  
  family = NULL,  
  color = NULL,  
  colour,  
  ...  
)
```

cowplot 小结

- `plot_grid`: 按 grid 安排
- `ggdraw()` + `draw_plot`: free style arrangement

use `gridExtra::grid.arrange`

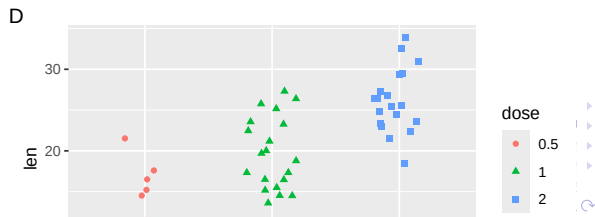
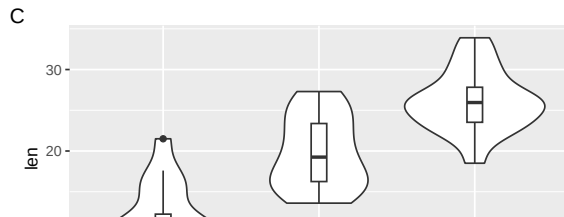
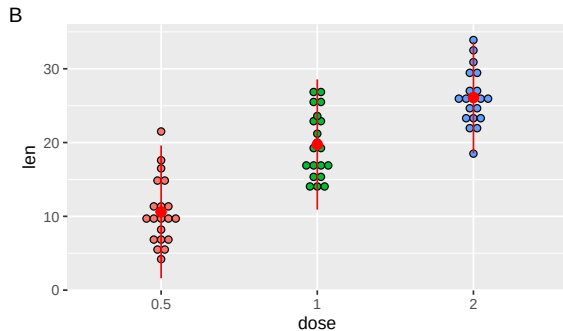
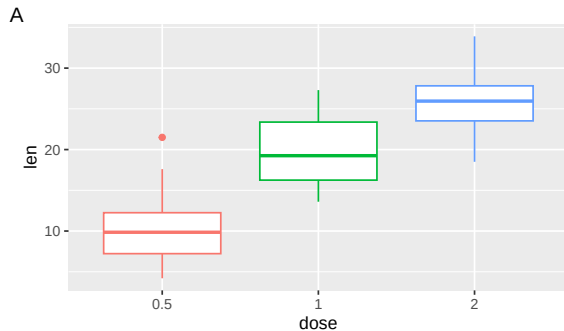
Create four plots, again

```
library(ggplot2); library("gridExtra");
df <- ToothGrowth
df$dose <- as.factor(df$dose)

bp <- ggplot(df, aes(x=dose, y=len, color=dose)) +
  geom_boxplot() +
  theme(legend.position = "none") + labs( tag = "A");
dp <- ggplot(df, aes(x=dose, y=len, fill=dose)) +
  geom_dotplot(binaxis='y', stackdir='center')+
  stat_summary(fun.data=mean_sdl, mult=1,
               geom="pointrange", color="red")+
  theme(legend.position = "none") + labs( tag = "B")
vp <- ggplot(df, aes(x=factor(dose), y=len)) +
  geom_violin()+
  geom_boxplot(width=0.1) + labs( tag = "C")
sc <- ggplot(df, aes(x=dose, y=len, color=dose, shape=dose)) +
  geom_jitter(position=position_jitter(0.2))+
  theme(legend.position = "none") +
  theme_gray() + labs( tag = "D")
```

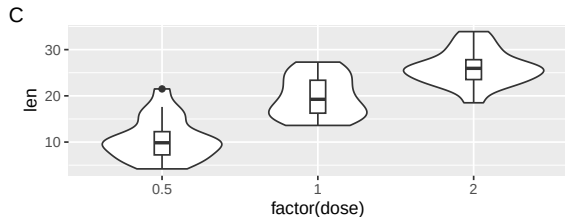
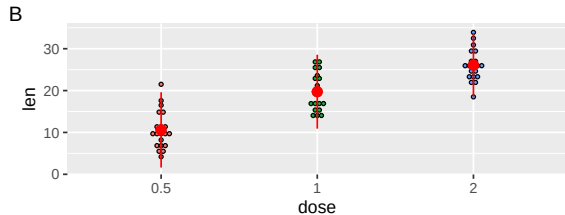
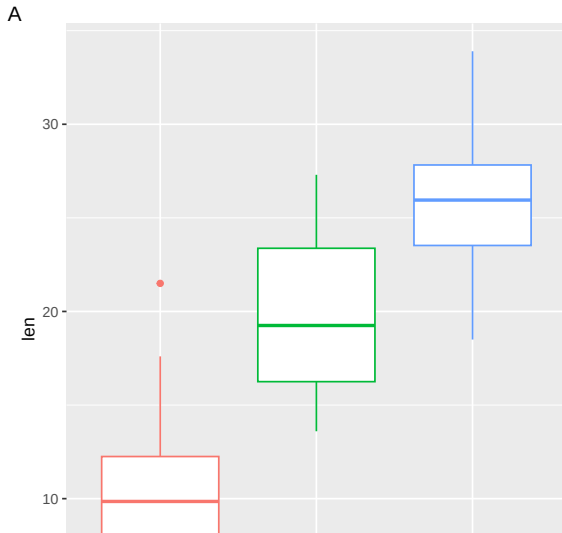
use `gridExtra::grid.arrange`

```
grid.arrange(bp, dp, vp, sc, ncol=2, nrow=2);
```



use layout_matrix parameter in grid.arrange

```
grid.arrange(bp, dp, vp, sc, ncol = 2,
             layout_matrix = cbind(c(1,1,1), c(2,3,4)));
```



explain layout_matrix

```
grid.arrange(bp, dp, vp, sc,
              ncol = 2, ## two columns
              layout_matrix = cbind(c(1,1,1), c(2,3,4)) ## specify the layout
              );
```

How the layout look like??

```
cbind(c(1,1,1), c(2,3,4));
```

	[,1]	[,2]
[1,]	1	2
[2,]	1	3
[3,]	1	4

make a different layout???

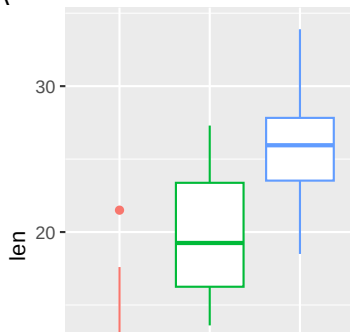
three columns, A and B take the first two, C and D take the third one.

```
( laymat = cbind(c(1,1), c(2,2), c(3,4)) );
```

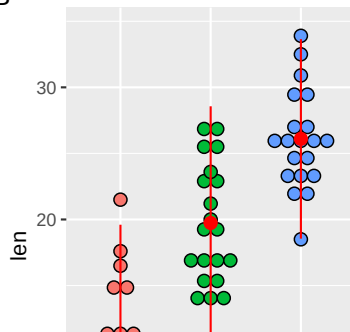
	[,1]	[,2]	[,3]
[1,]	1	2	3
[2,]	1	2	4

```
grid.arrange(bp, dp, vp, sc, ncol = 3, layout_matrix = laymat);
```

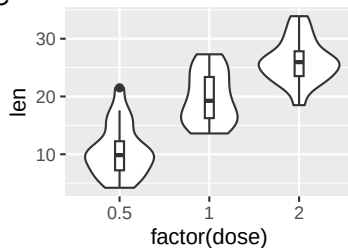
A



B



C



D



Extended reading

Explore the grid package

- create graphical objects (grobs)
- arrange multiple grobs using `arrangeGrob` function

Explore the gridExtra package

- plot table
- ...

Other ggplot2 extensions

See the gallery at <https://exts.ggplot2.tidyverse.org/gallery/>. Or Google ggplot2 extensions gallery.

complex layout 小结

Essentials for combining multiple graphs in one:

- ordering
- layout

cowplot

- plot_grid
- draw_plot

gridExtra

- grid.arrange

Section 3

ggplot2 进阶 2: 如何写公式?

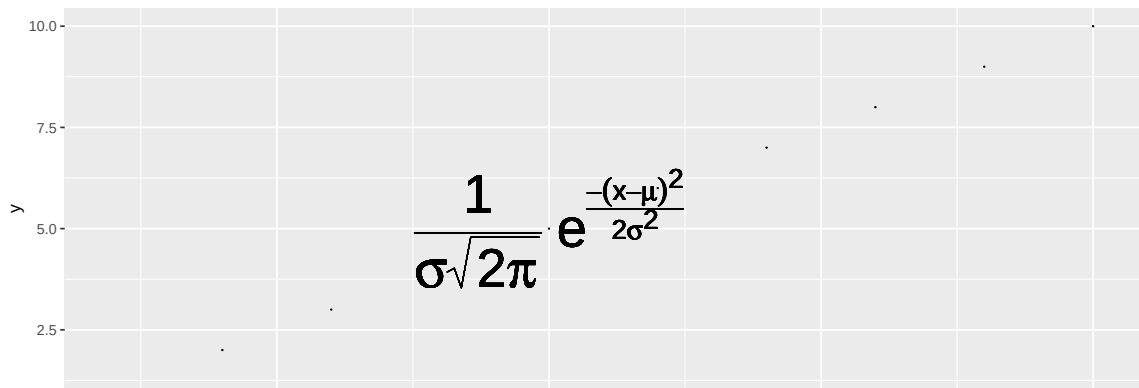
two kinds of equations

- ① expression w/o variable substitution
- ② expression w/ variable substitution

第一种写法: **expression**, 没有数值代入

```
eq <- expression(paste(frac(1, sigma*sqrt(2*pi)), " ",
                        plain(e)^{frac(-(x-mu)^2, 2*sigma^2)}));

ggplot( data.frame(x=1:10, y=1:10), aes( x,y ) ) +
  geom_point( size = 0 ) +
  geom_text(data = NULL, x = 5, y = 5, size = 12,
            label = as.character(eq), parse = TRUE );
```



公式中的写法之代数符号

分类	R 的表达式	显示结果
代数符号	<code>expression(x + y)</code>	$x + y$
	<code>expression(x - y)</code>	$x - y$
	<code>expression(x * y)</code>	xy
	<code>expression(x / y)</code>	x/y
	<code>expression(x %+-% y)</code>	$x \pm y$
	<code>expression(x %/% y)</code>	$x \div y$
	<code>expression(x %*% y)</code>	$x \times y$
	<code>expression(x %.% y)</code>	$x \cdot y$
	<code>expression(x[i])</code>	x_i
	<code>expression(x^2)</code>	x^2
	<code>expression(sqrt(x))</code>	$\sqrt{x}$
	<code>expression(sqrt(x,y))</code>	$\sqrt[y]{x}$
	<code>expression(list(x,yz))</code>	x, y, z

... 更多, 不在此介绍了。

希腊字符

代码

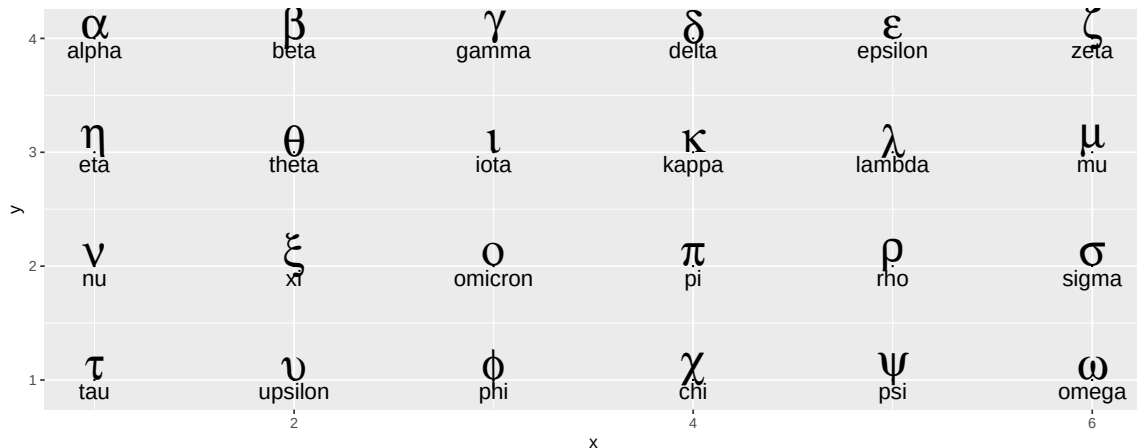
```
library(ggplot2);
greeks <- c("Alpha", "Beta", "Gamma", "Delta", "Epsilon", "Zeta",
            "Eta", "Theta", "Iota", "Kappa", "Lambda", "Mu",
            "Nu", "Xi", "Omicron", "Pi", "Rho", "Sigma",
            "Tau", "Upsilon", "Phi", "Chi", "Psi", "Omega");

dat <- data.frame( x = rep( 1:6, 4 ), y = rep( 4:1, each = 6), greek = greeks );

plot2 <-
  ggplot( dat, aes(x=x,y=y) ) + geom_point(size = 0) +
  # 画希腊字符, 注意下面两行代码的区别
  geom_text( aes( x, y + 0.1, label = tolower( greek ) ), size = 10, parse = T ) +
  geom_text( aes( x, y - 0.1, label = tolower( greek ) ), size = 5 );
```

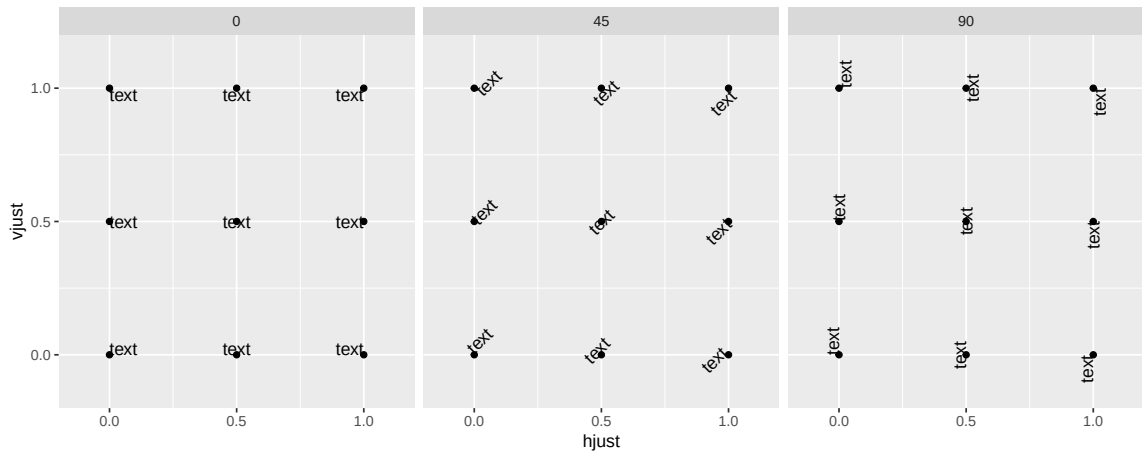


希腊字符, cont.



hjust 和 vjust

`geom_text( aes( angle, hjust, vjust ) )` 三参数详解



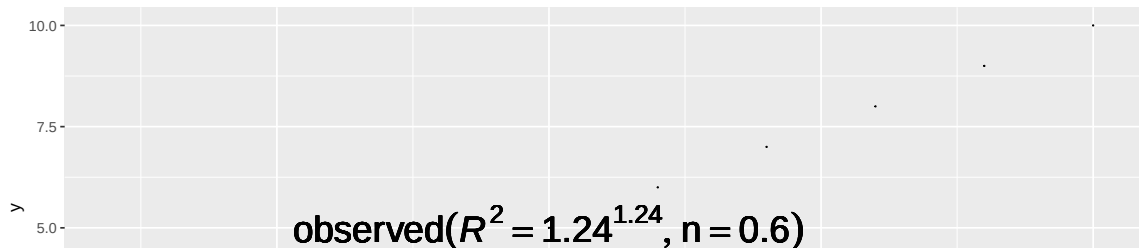
第二种写法: 有变量代入值的公式

举例 1:

```
x <- 1.24;
y <- 0.6;

ex <- bquote(.(parse(text=paste( "observed (", "italic(R)^2==",
                                x,  "^bold(", x,  ")", n == ", y,  ")",
                                sep = "  " )))) );

ggplot( data.frame(x=1:10, y=1:10), aes( x,y ) ) + geom_point( size = 0 ) +
  geom_text(data = NULL, x = 5, y = 5, size = 8,
            label = as.character(ex), parse = TRUE );
```



有代入变量值的公式举例 2:

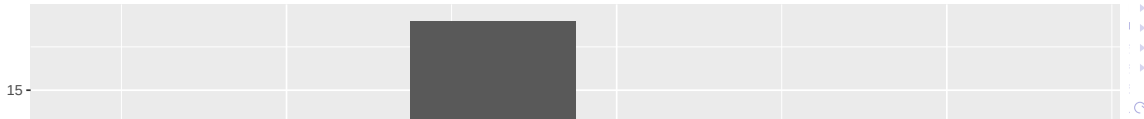
使用 paste 和 substitute :

```
x_mean <- 1.5;
x_sd <- 1.2;

# 表达式
ex <- substitute(
  paste(X[i], " ~ N(", mu, "=", m, ", ", sigma^2, "=", s2, ")"),
  list(m = x_mean, s2 = x_sd^2)
);

# histogram
ggplot( data.frame( x = rnorm(100, x_mean, x_sd) ), aes( x ) ) +
  geom_histogram( binwidth=0.5 ) +
  ggtitle(ex); ## 为什么不需要 parse = TRUE ???
```

$$X_i \sim N(\mu=1.5, \sigma^2=1.44)$$

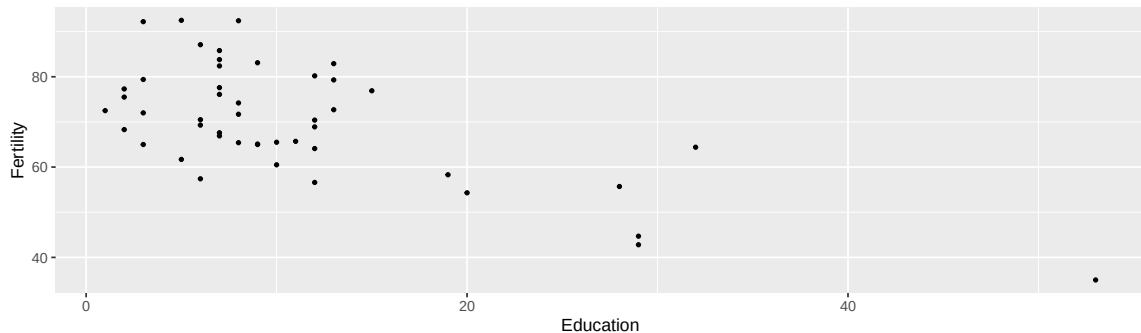


详解

示例: 显示两组数据间的相关性

作图

```
ggplot( swiss, aes( x = Education, y = Fertility ) ) +  
  geom_point( shape = 20 );
```

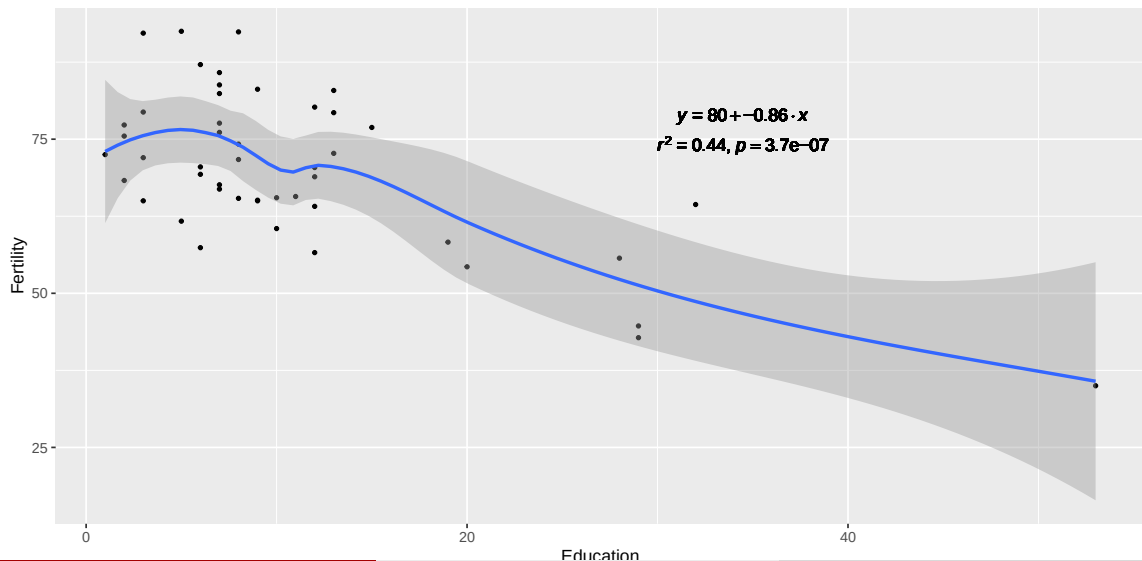


分析

```
with( swiss, cor.test( Education, Fertility )$estimate );
```

在图中加入公式和统计信息

先展示一下结果



公式详解

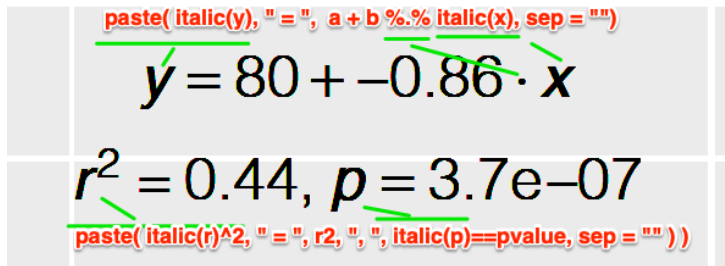


Figure 1: equation explained!

公式详解, cont.

以下代码实现两个任务:

- ① 将两个公式上下放置 atop (<equation_1> , <equation_2>);
- ② 将公式中的某些值替换为数值 substitute(<equation>, list(...))

```
## 计算 ...
m = lm(Fertility ~ Education, swiss);
c = cor.test( swiss$Fertility, swiss$Education );

## 生成公式
eq <- substitute( atop( paste( italic(y), " = ", a + b %.% italic(x), sep = ""),
                             paste( italic(r)^2, " = ", r2, " ", italic(p) == pvalue, sep = "" ) ),
                  list(a = as.vector( format(coef(m)[1], digits = 2) ),
                      b = as.vector( format(coef(m)[2], digits = 2) ),
                      r2 = as.vector( format(summary(m)$r.squared, digits = 2) ),
                      pvalue = as.vector( format( c$p.value , digits = 2) ) )
);

## 用 as.expression 对公式进行转化
eq <- as.character(as.expression(eq));
```

作图代码

```
## 作图, 三个图层; 特别是 geom_text 使用自己的 data 和 aes ...
ggplot(swiss, aes( x = Education, y = Fertility )) +
  geom_point( shape = 20 ) +
  geom_smooth( se = T ) + ## smooth line ...
  geom_text( data = NULL,
             aes( x = 30, y = 80, label= eq, hjust = 0, vjust = 1), ## hjust, vjust ???
             size = 4, parse = TRUE, inherit.aes=FALSE); ## 注意: parse = TRUE !!!
```

公式详解

italic(y) == a + b %.% italic(x)*

$$y = 80 + -0.86 \cdot x$$

*italic(r)^2 ~ "=" ~ r^2 * , " ~ italic(p) == pvalue*

$$r^2 = 0.44, p = 3.7e-07$$

等号及两边的空格 逗号, 左边没空格, 右边有 - 另一种等号的写法 默认两边都是空格

Figure 2: equation explained!

注

- 引号两边必须有 * 或 ~ 字符, ~ 表示空格, * 表示什么都没有。~~ 表示两个空格。

公式小结

- 表达式
- 罗马字符
- 代入变量

Section 4

ggplot2 进阶 3: 核心在于先计算再做图

举例说明

先看数据 (来自 talk05):

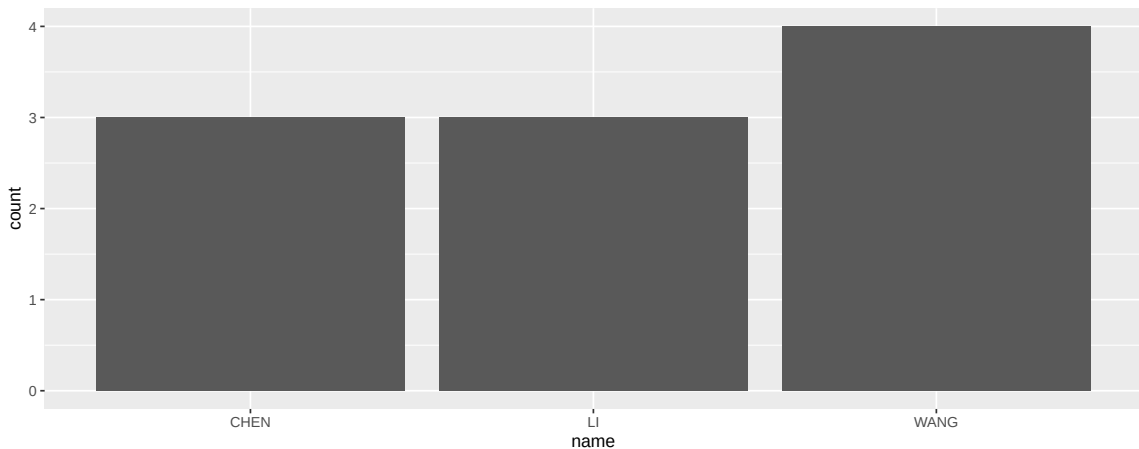
```
grades2 <- read_delim( file = "data/talk05/grades2.txt", delim = "\t",
                        quote = "", col_names = T);
knitr::kable( grades2 );
```

name	class	course	grade
CHEN	1	bioinformatics	90
CHEN	1	chemistry	92
CHEN	2	chinese	35
CHEN	3	german	62
LI	1	bioinformatics	44
LI	2	chinese	68
LI	3	microbiology	95
LI	3	japanese	90
WANG	1	bioinformatics	35
WANG	1	chemistry	76
WANG	1	mathmatics	82
WANG	3	german	100
WANG	3	spanish	78

geom_bar

任务: 画出每位学生及格的课程数

```
ggplot( grades2 %>% filter( grade >= 60 ), aes( name ) ) +  
  geom_bar();
```

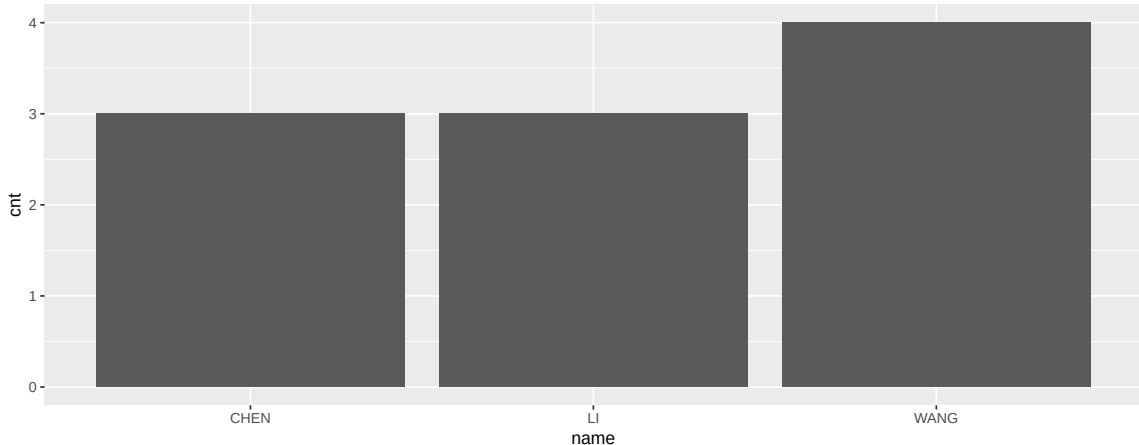


geom_bar , cont.

以上命令, 实际上等于:

先做统计

```
cnt <- grades2 %>% group_by( name ) %>% summarise( cnt = sum( grade >= 60 ) );  
ggplot( cnt, aes( x = name, y = cnt ) ) + geom_bar( stat = "identity" );
```



default stat behaviors (默认计算方法)

- `geom_bar` : `count`
- `geom_boxplot` : `boxplot`
- `geom_count` : `sum`
- `geom_density` : `density`
- `geom_histogram` : `bin`
- `geom_quantile` : `quantile ...`

Section 5

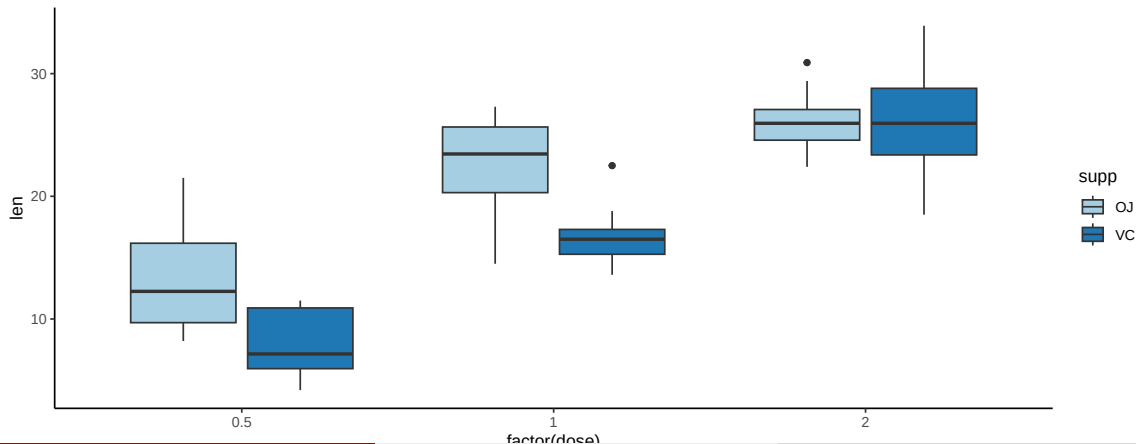
ggplot2 进阶 4: themes and legends

改变 theme()

theme 调整 包括 theme() 函数, 用于调整各个 elements

和 theme_xxx() 函数, 直接使用已经定制好的内容;

```
ggplot(ToothGrowth, aes(x=factor( dose ), y=len, fill=supp)) +  
  geom_boxplot() + scale_fill_brewer( palette = "Paired" ) + theme_classic();
```



ggplot2 中的主题

- `theme_gray` : 系统默认主题
- `theme_bw` , `theme_linedraw`, `theme_light`, `theme_dark`, `theme_minimal` , `theme_classic`, `theme_void()`

see here for a complete list: <https://ggplot2.tidyverse.org/reference/ggtheme.html>

theme() 函数

除了 theme_ 用于调整整体视觉效果外, ggplot2 还提供了 theme() 函数用于细调。

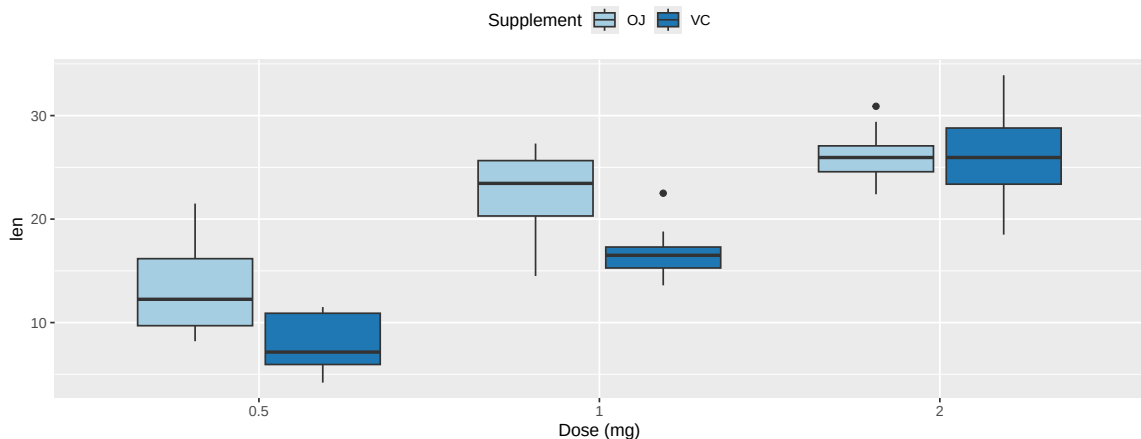
- line, rect, text, title : 整体框架
- axis.<compoment> : 调整坐标轴
- legend.<parameter> : 调整图例
- plot.<> : 控制 title, subtitle 等细节
- panel.<...> : 调整 facet 情况下的 panel (facet 下面会介绍)
- strip.<...> : 调整 facet 的 标题细节 ...

更多详见:

官方: <https://ggplot2.tidyverse.org/reference/theme.html>

legend 细调

```
ggplot(ToothGrowth, aes(x=factor( dose ), y=len, fill=supp)) +
  geom_boxplot() + scale_fill_brewer( palette = "Paired" ) +
  labs( fill = "Supplement", x = "Dose (mg)" ) +
  theme( legend.position = "top" )
```



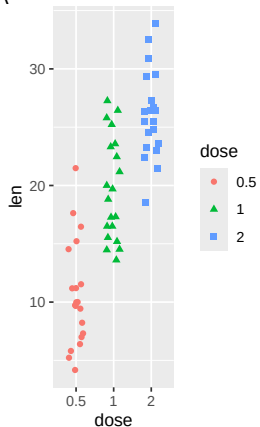
labs() function: Modify axis, legend, and plot labels

```
labs(  
  ...,  
  x = "<x label>",  
  y = "<y label>",  
  colour = "<legend title>", # 与 aes 里的 colour 配合使用  
  fill = "<legend title>", # 与 aes 里的 fill 配合使用  
  shape = "<legend title>", # 与 aes 里的 shape 配合使用  
  title = waiver(),  
  subtitle = waiver(),  
  caption = waiver(),  
  tag = waiver(),  
  alt = waiver(),  
  alt_insight = waiver()  
)
```

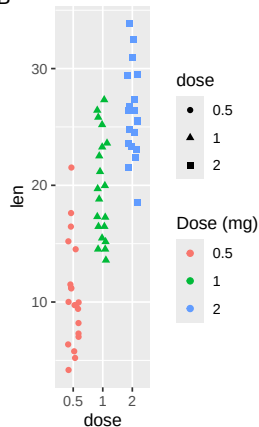
labs() with examples

```
grid.arrange(sc + labs(tag = "A"),
  sc + labs( colour = "Dose (mg)" , tag = "B"),
  sc + labs( shape = "Dose (mg)" , tag = "C"),
  sc + labs( colour = "Dose (mg)", shape = "Dose (mg)", tag = "D" ),
  ncol=4, nrow =1);
```

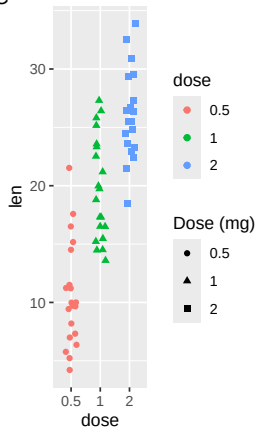
A



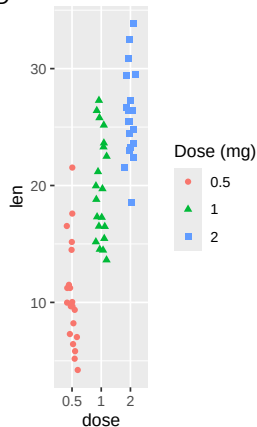
B



C



D



Section 6

ggplot2 进阶 5: stacked bars 及其它

stacked bars

应用场景：宏基因组多样样本物种丰度图

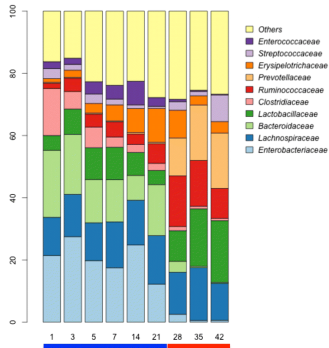


Figure 3: Microbiome 3, 28 2015

stacked bars , cont.

load data

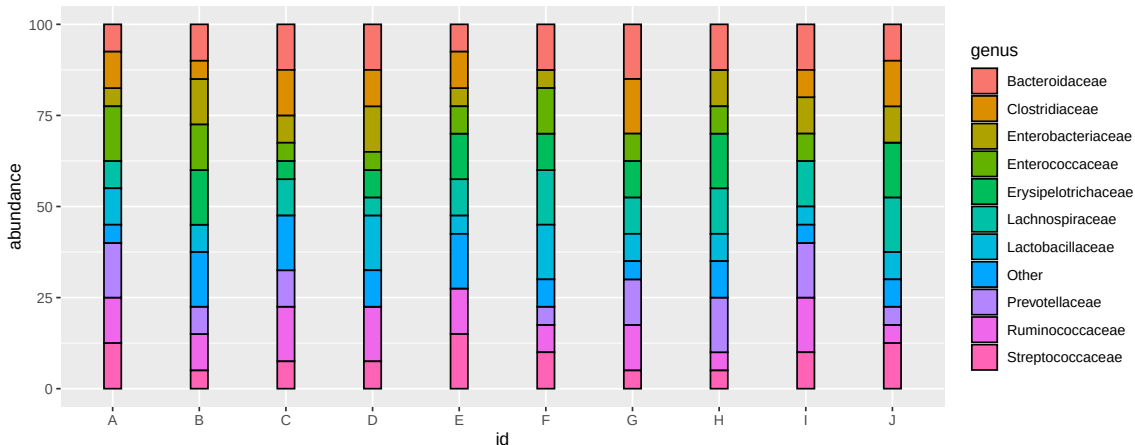
```
speabu <-read_tsv( file = "data/talk09/mock_species_abundance.txt" );
head( speabu );
```

A tibble: 6 x 3

	id	genus	abundance
	<chr>	<chr>	<dbl>
1	A	Enterobacteriaceae	5
2	A	Lachnospiraceae	7.5
3	A	Bacteroidaceae	7.5
4	A	Lactobacillaceae	10
5	A	Clostridiaceae	10
6	A	Ruminococcaceae	12.5

stacked bars , cont.

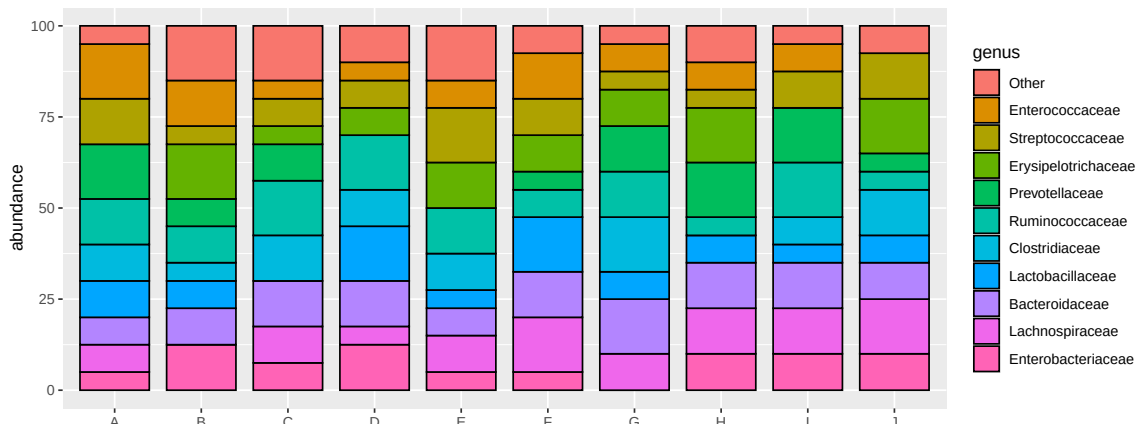
```
ggplot( speabu, aes( x = id, y = abundance, fill = genus ) ) +
  geom_bar( stat = "identity", position = "stack", color = "black", width = 0.2 );
```



指定 Genus 展示顺序

factor 的操纵详见第 4 章。

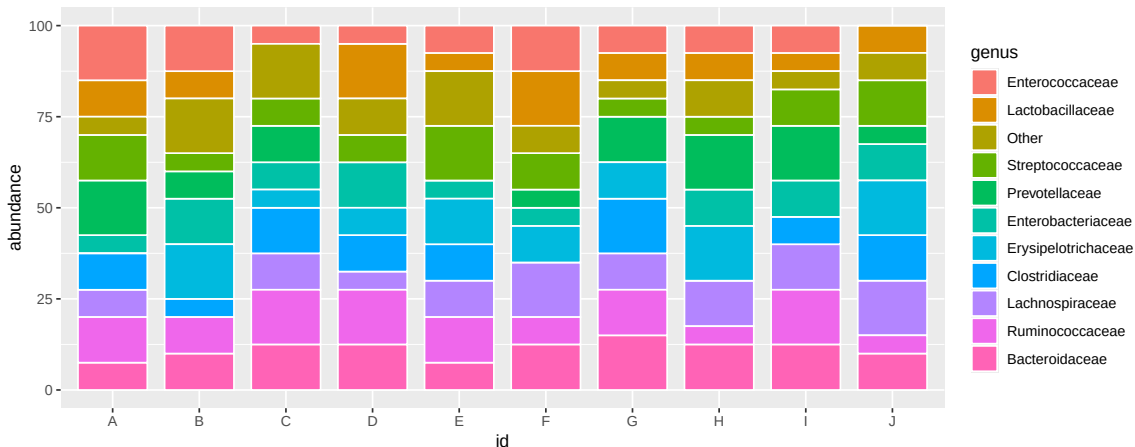
```
speabu$genus <- factor( speabu$genus, levels = rev( c( "Enterobacteriaceae", "Lachnospiraceae",
  "Clostridiaceae", "Ruminococcaceae", "Prevotellaceae", "Erysipelotrichaceae", "Streptococcaceae",
  "Enterococcaceae", "Other" ) ) )
ggplot( speabu, aes( x = id, y = abundance, fill = genus ) ) +
  geom_bar( stat = "identity", position = "stack", color = "black", width = 0.8 );
```



按丰度排序

按丰度中值大小排序

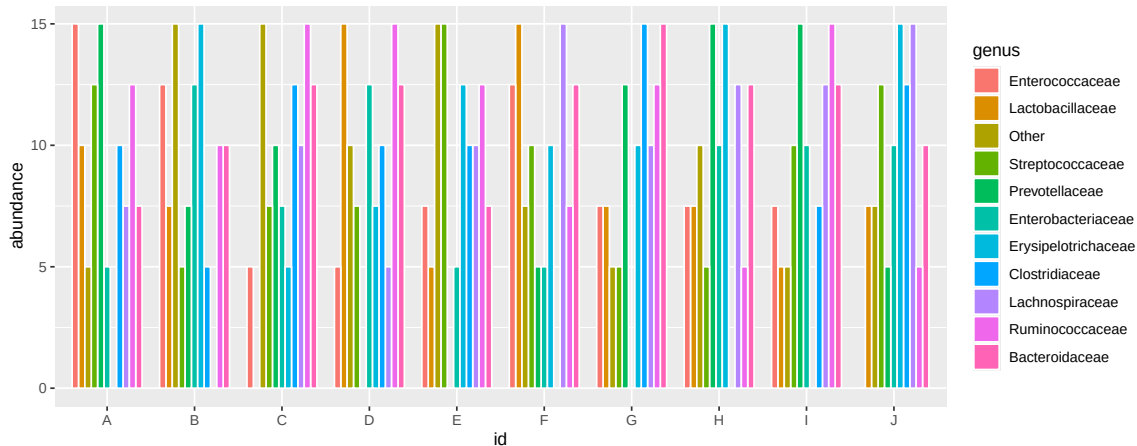
```
speabu$genus <- reorder( speabu$genus, speabu$abundance, median );
ggplot( speabu, aes( x = id, y = abundance, fill = genus ) ) +
  geom_bar( stat = "identity", position = "stack", color = "white", width = 0.8 );
```



position = "stack" 又是什么??

position = "dodge" : plot bars next to each other ...

```
ggplot( speabu, aes( x = id, y = abundance, fill = genus ) ) +  
  geom_bar( stat = "identity", position = "dodge", color = "white", width = 0.8 );
```



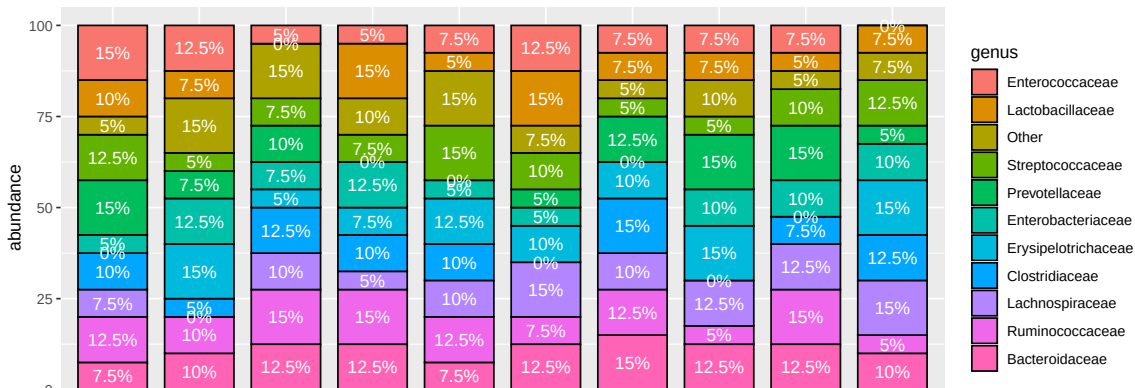
显示数值 ...

```
## 先计算显示位置
```

```
speabu <- speabu %>% arrange( id, desc( factor( genus ) ) ) %>%
  group_by( id ) %>% mutate( ypos = cumsum( abundance ) - abundance / 2 );
```

```
## 画图
```

```
ggplot( speabu, aes( x = id, y = abundance, fill = genus ) ) +
  geom_bar( stat = "identity", position = "stack", color = "black", width = 0.8 ) +
  geom_text( aes( y = ypos, label = paste( abundance, "%", sep = "" ) ), color = "white" );
```



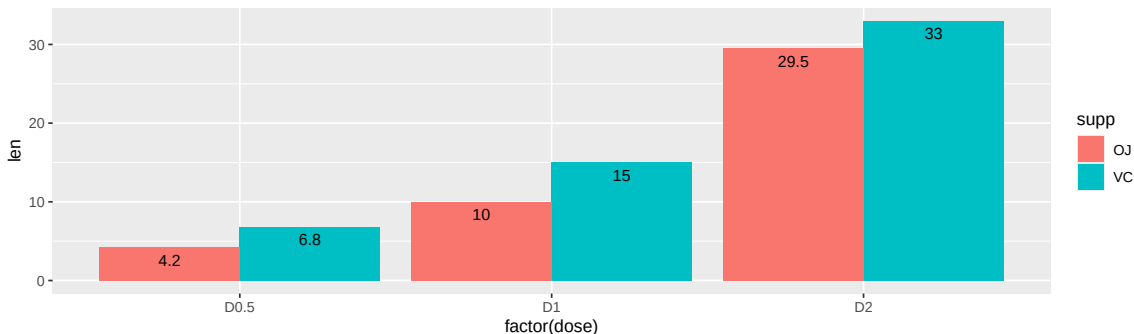
显示数值 ... , cont.

要点

- 使用 `ddplyr` 的 `cumsum()` 函数 ...
- 计算位置: 当前累加值 - 自身值/2, 使数字显示在当前值的中间
- 累加前, 要对数据按 factors 进行排序; 通过 `arrange` 函数实现;

在 position = "dodge" 的情况下添加 label

```
df2 <- data.frame(supp=rep(c("VC", "OJ"), each=3),
                  dose=rep(c("D0.5", "D1", "D2"),2),
                  len=c(6.8, 15, 33, 4.2, 10, 29.5))
ggplot( df2, aes(x=factor(dose), y=len, fill=supp)) +
  geom_bar(stat="identity", position=position_dodge())+
  geom_text(aes(label=len), vjust=1.6, color="black",
            position = position_dodge(0.9), size=3.5)
```



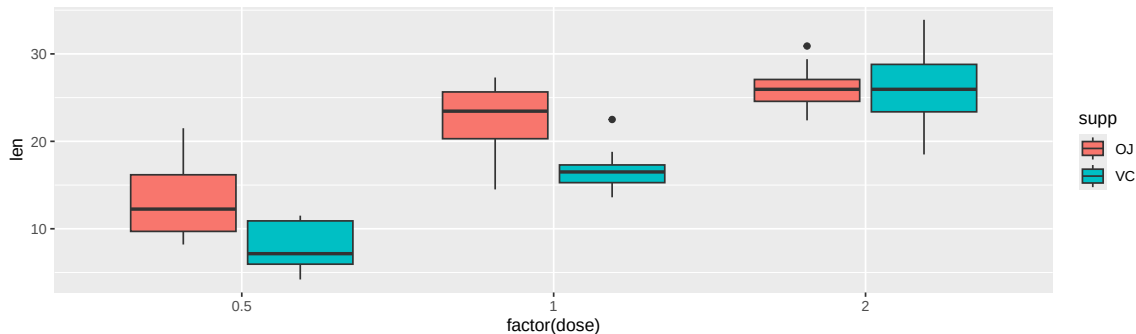
position 的其它取值

除了 “dodge”, “stack” 之外, position 还可以:

- `position = position_stack(reverse = TRUE)`
- `position = position_dodge(reverse = TRUE)`
- `position = position_identity()`
- `position = position_jitter()` : jitter points to avoid overplotting ...
- `position = position_nudge()` : is generally useful for adjusting the position of items on discrete scales by a small amount

不同的图层有不同默认值

```
ggplot(ToothGrowth, aes(x=factor( dose ), y=len, fill=supp)) +  
  geom_boxplot()
```



`geom_boxplot()` : 默认为 `dodge`

more to read

ggplot2 的在线书

<https://ggplot2-book.org/themes>

Section 7

Exercise and home work

总结, 本节内容

ggplot2 基础

- 优缺点
- 用法
- 基本组成

ggplot2 进阶

- 颜色和色板
- 复杂 layout 的实现
- 公式
- ggplot2 的数据统计逻辑

更多阅读

- Ggplot2: Elegant Graphics for Data Analysis, Book by Hadley Wickham
- ggplot2 gallery provided by RStudio on Github

写在后面

- ① ggplot2 博大精深，需要一门课去讲
- ② 上手容易，精通难
- ③ 太多记忆点
- ④ 本节内容只涉及了基础中的基础，更多内容，包括进阶技巧和生信相关的扩展包，更多的需要同学们自行探索
- ⑤ 遇到不会的图，先百度/Google，找包和代码

下次预告

data summary and modeling

作业

- Exercises and homework 目录下 talk09-homework.Rmd 文件;
- 完成时间: 见钉群的要求